Agentic Manufacturing Pipeline — Architecture

Demand → Analytics, 12 agents, one fixed sequence. Built in `agentic/`, fully isolated from the working `r` platform.

✓ Confirmed: Python LangGraph (not LangGraph.js) + FastAPI

- Single-shot structured output
- ReAct loop (reason → act → observe)
- Calls an existing deterministic service
- Persisted state

USER-FACING LAYER

`agentic/web` — Next.js, unchanged framework, new pages

Planner Dashboard

`/agentic-pipeline` · role: PLANNER — trigger a run, view stage outputs, edit & replay from any stage.

Admin Dashboard

`/agentic-pipeline/admin` · role: ADMIN/OWNER — full raw trace, every stage, every retry, across all runs.

↓ HTTP

ORCHESTRATOR — PYTHON

`services/agent-orchestrator` — FastAPI + LangGraph (Python), one `StateGraph`, 12 nodes, fixed order

START HERE

01

Demand

SINGLE-SHOT

Forecast from raw signals. Nothing upstream — easiest to sanity-check by eye.

02

S&OP

SINGLE-SHOT

Reconcile demand against high-level capacity.

03

MPS

SINGLE-SHOT

Convert plan into SKU/period quantities.

04

BOM

05

MRP

06

Procurement

SINGLE-SHOT

Validate/fetch component structure.

↳ routing-service

REACT LOOP

Check inventory → *decide* whether to also check alternate suppliers, based on the result.

↳ inventory-service

SINGLE-SHOT

Draft purchase recommendations. Commits real spend.

07

APS Scheduling

SINGLE-SHOT

LLM explains the result — doesn't invent the schedule.

↳ scheduling-service (real engine)

08

Shop Floor

SINGLE-SHOT

Reports ground-truth execution status.

09

Quality

SINGLE-SHOT

Inspection outcome, pass/fail/rework.

10

Warehouse

SINGLE-SHOT

Putaway / allocation.

11

Dispatch

SINGLE-SHOT

Logistics / delivery execution.

12

Analytics

SINGLE-SHOT

Retrospective — nothing to approve.

🧠 Every node above calls out to **Groq** (free tier, e.g. `llama-3.3-70b-versatile`) for its reasoning — output forced through a **Pydantic schema**, one retry on validation failure.

↓ writes every stage, unconditionally

DATA LAYER

Postgres — `aps_platform_agentic`

AGENTPIPELINERUN	
`id, status, currentStage, createdAt`	mirrors ScheduleRun
AGENTSTAGERESULT	
`runId, stage, agentOutput`	`jsonb`
`humanEdited, editedByUserId`	`nullable`

ALREADY BUILT & TESTED

Deterministic services (called as tools, not reinvented)

scheduling-service

Forward/Backward/Bottleneck engine, capacity buckets, breakdown & fence logic — all verified this session.

inventory-service / routing-service

A row is written for **every** stage of **every** run — this table is the full trace, not just an approval record.

Real stock levels, BOM structure — read-only tool calls from MRP/BOM agents.

agentic/PLAN.md — full written rationale for every decision on this diagram